CONTENTS

4 Tests of Hypotheses and Confidence Intervals

4.1 Test of Hypotheses

Consider the linear model with

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} \text{ and } V(\mathbf{Y}) = \Sigma$$

This can also be expressed as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $E(\boldsymbol{\epsilon}) = \mathbf{0}$ and $V(\boldsymbol{\epsilon}) = \Sigma$.

Typical null hypothesis (H_0)

- is a status quo or prevailing viewpoint about a population
- specifies the values for one or more elements of $\boldsymbol{\beta}$
- specifies the values for some linear functions of the elements of $\boldsymbol{\beta}$

An alternative hypothesis (H_1)

- is an alternative to the null hypothesis – the change in the population that the researcher hopes is true

We may test

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d} \quad \text{vs} \quad H_1 : \mathbf{C}\boldsymbol{\beta} \neq \mathbf{d}$$

where

- $\mathbf{C}$ is an $m \times k$ matrix of constants
- $\mathbf{d}$ is an $m \times 1$ vector of constants

The null hypothesis is rejected if it is shown to be sufficiently incompatible with the observed data.

Failing to reject H_0 is **not** the same as proving H_0 is true.

- too little data to accurately estimate $\mathbf{C}\boldsymbol{\beta}$
- relatively large variation in $\boldsymbol{\epsilon}$ (or $\mathbf{Y}$)
- if $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$ is false, $\mathbf{C}\boldsymbol{\beta} - \mathbf{d}$ may be "small"

You can never be completely sure that you made the correct decision

- Type I error (significance level):
  $P(H_0 \text{ is rejected}|H_0 \text{ is true})$
- Type II error:
  $P(H_0 \text{ is rejected}|H_0 \text{ is true})$

Basic considerations in specifying a null hypothesis $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$

(i) $\mathbf{C}\boldsymbol{\beta}$ should be estimable.

(ii) Inconsistencies should be avoided, i.e., $\mathbf{C}\boldsymbol{\beta} = \mathbf{d}$ should be a consistent set of equations

(iii) Redundancies should be eliminated, i.e., in $\mathbf{C}\boldsymbol{\beta} = \mathbf{d}$ we should have

$$\text{rank}(\mathbf{C}) = \text{number-of-rows-in-}\mathbf{C}$$

4.2 Hypothesis Tests for Estimable Function

Consider the following effects models:

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij} \quad \begin{array}{l} i = 1, 2, 3 \\ j = 1, \ldots, n_i \end{array}$$

In this case

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{31} \\ Y_{32} \\ Y_{33} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} + \begin{bmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{21} \\ \epsilon_{31} \\ \epsilon_{32} \\ \epsilon_{33} \end{bmatrix}$$

4.2.1 The Mean Response For Any Treatments

By definition

$$E(Y_{ij}) = \mu + \alpha_i \text{ is estimable.}$$

We can test

$$H_0 : \mu + \alpha_1 = 60 \text{ seconds}$$

against

$$H_1 : \mu + \alpha_1 \neq 60 \text{ seconds}$$

(two-sided alternative)

Or we can test

$$H_0 : \mu + \alpha_1 = 60 \text{ seconds}$$

against

$$H_1 : \mu + \alpha_1 < 60 \text{ seconds}$$

(one-sided alternative)

In this case

$$\mu + \alpha_1 = \mathbf{c}^T \boldsymbol{\beta} \quad \text{where} \quad \mathbf{c} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Note that this quantity is estimable, i. e.,

$$\mathbf{c}^T\boldsymbol{\beta} = \mu + \alpha_1 = E\left[\begin{pmatrix}\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0\end{pmatrix}\mathbf{Y}\right].$$

Then, any solution

$$\mathbf{b} = (\mathbf{X}^T\Sigma^{-1}\mathbf{X})^-\mathbf{X}^T\Sigma^{-1}\mathbf{Y}$$

to the generalized least squares estimating equations

$$\mathbf{X}^T\Sigma^{-1}\mathbf{X}\mathbf{b} = \mathbf{X}^T\Sigma^{-1}\mathbf{Y}$$

yields the **same value** for $\mathbf{c}^T\mathbf{b}$ and it is the **unique blue** for $\mathbf{c}^T\boldsymbol{\beta}$.

We will reject $H_0 : \mathbf{c}^T\boldsymbol{\beta} = 60$ if

$$\mathbf{c}^T\mathbf{b} = \mathbf{c}^T(\mathbf{X}^T\Sigma^{-1}\mathbf{X})^-\mathbf{X}^T\Sigma^{-1}\mathbf{Y}$$

is too far away from 60.

If $\mathrm{V}(\mathbf{Y}) = \sigma^2 I$, then any solution

$$\mathbf{b} = (\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}$$

to the least squares estimating equations

$$\mathbf{X}^T\mathbf{X}\mathbf{b} = \mathbf{X}^T\mathbf{Y}$$

yields the **same value** for $\mathbf{c}^T\mathbf{b}$, and $\mathbf{c}^T\mathbf{b}$ is the **unique blue** for $\mathbf{c}^T\boldsymbol{\beta}$.

We will reject $H_0 : \mathbf{c}^T\boldsymbol{\beta} = 60$ if

$$\mathbf{c}^T\mathbf{b} = \mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}$$

is too far away from 60.

4.2.2 Difference between the mean response for two treatments

$$\alpha_1 - \alpha_3 = (\mu + \alpha_1) - (\mu + \alpha_3)$$
$$= \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & \frac{-1}{3} & \frac{-1}{3} & \frac{-1}{3} \end{pmatrix} E(\mathbf{Y})$$

and we can test

$$H_0 : \alpha_1 - \alpha_3 = 0 \quad \text{vs.} \quad H_1 : \alpha_1 - \alpha_3 \neq 0$$

If $\mathrm{V}(\mathbf{Y}) = \sigma^2 I$, the unique blue for

$$\alpha_1 - \alpha_3 = (0\ 1\ 0\ \ -1)\boldsymbol{\beta} = \mathbf{c}^T\boldsymbol{\beta}$$

is

$$\mathbf{c}^T\mathbf{b} \text{ for any } \mathbf{b} = (\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}$$

Reject $H_0 : \alpha_1 - \alpha_3 = \mathbf{c}^T\boldsymbol{\beta} = 0$ if $\mathbf{c}^T\mathbf{b}$ is too far from 0.

4.2.3 Non Estimable Functions

It would not make much sense to attempt to test

$$H_0 : \alpha_1 = 3 \quad \text{vs.} \quad H_1 : \alpha_1 \neq 3$$

because $\alpha_1 = [0\ 1\ 0\ 0]\boldsymbol{\beta} = \mathbf{c}^T\boldsymbol{\beta}$ is not estimable.

- Although $E(Y_{1j}) = \mu + \alpha_1$ neither μ nor α_1 has a clear interpretation.
- Different solutions to the normal equations produce different values for
$$\hat{\alpha}_1 = \mathbf{c}^T\mathbf{b} = \mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}$$
- To make a statement about α_1, an additional restriction must be imposed on the parameters in the model to give α_1 a precise meaning.

4.3 Consistencies and Redundancies

For $\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$, consider testing

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \begin{bmatrix} -3 \\ 60 \\ 70 \end{bmatrix} \text{ vs. } H_1 : \mathbf{C}\boldsymbol{\beta} \neq \begin{bmatrix} -3 \\ 60 \\ 70 \end{bmatrix}$$

In this case $\mathbf{C}\boldsymbol{\beta}$ is estimable, but there is an inconsistency. If the null hypothesis is true,

$$\mathbf{C}\boldsymbol{\beta} = \begin{bmatrix} \alpha_1 - \alpha_3 \\ \mu + \alpha_1 \\ \mu + \alpha_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 60 \\ 70 \end{bmatrix}$$

Then $\mu + \alpha_1 = 60$ and $\mu + \alpha_3 = 70$ implies

$$\begin{aligned}(\alpha_1 - \alpha_3) &= (\mu + \alpha_1) - (\mu + \alpha_3) \\ &= 60 - 70 \\ &= \textbf{-10}\end{aligned}$$

Such inconsistencies should be avoided.

For $\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$

consider testing

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \begin{bmatrix} -10 \\ 60 \\ 70 \end{bmatrix} \text{ vs. } H_1 : \mathbf{C}\boldsymbol{\beta} \neq \begin{bmatrix} -10 \\ 60 \\ 70 \end{bmatrix}$$

In this case $\mathbf{C}\boldsymbol{\beta}$ is estimable and the equations specified by the null hypothesis are consistent.

There is a redundancy

$$[1\ 1\ 0\ 0]\,\boldsymbol{\beta} = \mu + \alpha_1 = 60$$
$$[1\ 0\ 0\ 1]\,\boldsymbol{\beta} = \mu + \alpha_3 = 70$$

imply that

$$\begin{aligned}[0\ 1\ 0\ -1]\,\boldsymbol{\beta} &= \alpha_1 - \alpha_3 \\ &= (\mu + \alpha_1) - (\mu + \alpha_3) \\ &= 60 - 70 \\ &= -10\end{aligned}$$

The rows of $\mathbf{C}$ are not linearly independent, i.e., rank($\mathbf{C}$) < number of rows in $\mathbf{C}$.
There are many equivalent ways to remove a redundancy:

$$H_0 : \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \boldsymbol{\beta} = \begin{bmatrix} 60 \\ 70 \end{bmatrix}$$

$$H_0 : \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \boldsymbol{\beta} = \begin{bmatrix} -10 \\ 60 \end{bmatrix}$$

$$H_0 : \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \boldsymbol{\beta} = \begin{bmatrix} -10 \\ 70 \end{bmatrix}$$

$$H_0 : \begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 1 & 0 & 1 \end{bmatrix} \boldsymbol{\beta} = \begin{bmatrix} 50 \\ 130 \end{bmatrix}$$

are all equivalent.

In each case:

- The two rows of $\mathbf{C}$ are linearly independent and

  $$\begin{aligned} \text{rank}(\mathbf{C}) &= 2 \\ &= \text{number of rows in } \mathbf{C} \end{aligned}$$

- The two rows of $\mathbf{C}$ are a basis for the same 2-dimensional subspace of R^4.

  This is the 2-dimensional space spanned by the rows of

  $$\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

We will only consider null hypotheses of the form

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$$

where rank($\mathbf{C}$) = number of rows in $\mathbf{C}$. This leads to the following concept of a "testable" hypothesis.

4.4 Testable Hypothesis

Definition 1.
Consider a linear model

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$$

where

$$V(\mathbf{Y}) = \Sigma$$

and $\mathbf{X}$ is an $n \times k$ matrix. For an $m \times k$ matrix of constants $\mathbf{C}$ and an $m \times 1$ vector of constants $\mathbf{d}$, we will say that

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$$

is **testable** if

(i) $\mathbf{C}\boldsymbol{\beta}$ is estimable

(ii) rank$(\mathbf{C}) = m =$ number of rows in $\mathbf{C}$

To test $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$

(i) Use the data to estimate $\mathbf{C}\boldsymbol{\beta}$.

(ii) Reject $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$ if the estimate of $\mathbf{C}\boldsymbol{\beta}$ is to far away from $\mathbf{d}$.

- How much of the deviation of the estimate of $\mathbf{C}\boldsymbol{\beta}$ from $\mathbf{d}$ can be attributed to random errors?
- Need a probability distribution for the estimate of $\mathbf{C}\boldsymbol{\beta}$
- Need a probability distribution for a test statistic

4.5 Normal Theory Gauss-Markov Model

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \sim N(\mathbf{X}\boldsymbol{\beta},\ \sigma^2 I)$$

A least squares estimator $\mathbf{b}$ for $\boldsymbol{\beta}$ minimizes

$$(\mathbf{Y} - \mathbf{X}\mathbf{b})^T(\mathbf{Y} - \mathbf{X}\mathbf{b})$$

For any generalized inverse of $\mathbf{X}^T\mathbf{X}$,

$$\mathbf{b} = (\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T\mathbf{Y}$$

is a solution to the normal equations

$$(\mathbf{X}^T\mathbf{X})\mathbf{b} = \mathbf{X}^T\mathbf{Y}\ .$$

Result 1. Results for the Gauss-Markov model

For a testable null hypothesis

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$$

the OLS estimator for $\mathbf{C}\boldsymbol{\beta}$,

$$\mathbf{C}\mathbf{b} = \mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T\mathbf{Y}\ ,$$

has the following properties:

(i) Since $\mathbf{C}\boldsymbol{\beta}$ is estimable, $\mathbf{C}\mathbf{b}$ is invariant to the choice of $(\mathbf{X}^T\mathbf{X})^{-}$.

(ii) Since $\mathbf{C}\boldsymbol{\beta}$ is estimable, $\mathbf{C}\mathbf{b}$ is the unique BLUE for $\mathbf{C}\boldsymbol{\beta}$.

(iii) $E(\mathbf{C}\mathbf{b} - \mathbf{d}) = \mathbf{C}\boldsymbol{\beta} - \mathbf{d}$

Ans: $E(\mathbf{C}\mathbf{b} - \mathbf{d}) = E(\mathbf{C}\mathbf{b}) - \mathbf{d} = \mathbf{C}\boldsymbol{\beta} - \mathbf{d}$

(iv) $V(\mathbf{Cb}-\mathbf{d}) = V(\mathbf{Cb}) = \sigma^2\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T$

Ans:

$$\begin{aligned}
&V[\mathbf{Cb}-\mathbf{d}]\\
&= V[\mathbf{Cb}]\\
&= V[\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}]\\
&= \mathbf{C}(\mathbf{X}^T\mathbf{X})^-V(\mathbf{Y})\mathbf{X}[(\mathbf{X}^T\mathbf{X})^{-1}]^T\mathbf{C}^T\\
&= \sigma^2[\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{X}[\mathbf{X}^T\mathbf{X}]^T\mathbf{C}^T]
\end{aligned}$$

Since $\mathbf{C}\boldsymbol{\beta}$ is estimable, $\mathbf{C} = A\mathbf{X}$ for some A,

$$\begin{aligned}
V[\mathbf{Cb}] &= \sigma^2 A\mathbf{X}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{X}[(\mathbf{X}^T\mathbf{X})^-]^TA^T\\
&= \sigma^2 A\mathbf{X}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^TA^T\\
&= \sigma^2\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T
\end{aligned}$$

(v) $\mathbf{Cb}-\mathbf{d} \sim N(\mathbf{C}\boldsymbol{\beta}-\mathbf{d}, \sigma^2\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T)$

Ans:

Since $\mathbf{Cb}-\mathbf{d} = \mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}-\mathbf{d}$ is a linear combination of $\mathbf{Y}$, then $\mathbf{Cb}-\mathbf{d}$ follows a normal distribution with mean $\mathbf{C}\boldsymbol{\beta}-\mathbf{d}$ and variance $\sigma^2\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T$.

(vi) When $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$ is true,

$$\mathbf{Cb} - \mathbf{d} \sim N(\mathbf{0}, \sigma^2 \mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T)$$

Ans:

When H_0 is true, $\mathbf{C}\boldsymbol{\beta} - \mathbf{d} = 0$, thus $\mathbf{Cb} - \mathbf{d} \sim N(\mathbf{0}, \sigma^2 \mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T)$

(vii) Define

$$SS_{H_0} = (\mathbf{Cb}-\mathbf{d})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T]^{-1}(\mathbf{Cb}-\mathbf{d})$$

then

$$\frac{1}{\sigma^2}\mathrm{SS}_{H_0} \sim \chi^2_m(\lambda)$$

where $m = rank(\mathbf{C})$ and

$$\lambda = \frac{1}{\sigma^2}(\mathbf{C}\boldsymbol{\beta}-\mathbf{d})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T]^{-1}(\mathbf{C}\boldsymbol{\beta}-\mathbf{d})$$

Ans:

$\mathbf{Cb} - \mathbf{d} \sim N(\mathbf{C}\boldsymbol{\beta} - \mathbf{d}, \sigma^2\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T)$
Take $A = \frac{1}{\sigma^2}[\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T]^{-1}$ and
$\Sigma = \sigma^2\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T$
Clearly, $A\Sigma = I$ is idempotent.
Since $\mathbf{C}\boldsymbol{\beta}$ is estimable and $rank(\mathbf{C}) = m =$ number of rows in $\mathbf{C}$, then $\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T$ is positive definite and $(\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T)^{-1}$ exists.

$$\begin{aligned}\lambda &= \boldsymbol{\mu}^T A\boldsymbol{\mu}\\ &= \tfrac{1}{2\sigma^2}(\mathbf{C}\boldsymbol{\beta} - \mathbf{d})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T]^{-1}(\mathbf{C}\boldsymbol{\beta} - \mathbf{d})\end{aligned}$$

$\therefore \frac{SSH_0}{\sigma^2} \sim \chi^2_m(\lambda)$

(viii) $\frac{1}{\sigma^2}SS_{H_0} \sim \chi^2_m$ if and only if $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$ is true.

Ans:

Since $\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T > 0$, $\lambda > 0$ unless $(\mathbf{C}\boldsymbol{\beta} - \mathbf{d}) = \mathbf{0}$

Hence $\lambda = 0$ if and only if $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$ is true.

(ix) $E(SSE) = (n-k)\sigma^2$ where $k = \text{rank}(\mathbf{X}) = \text{rank}(\mathbf{P_X})$ and $n - k = \text{rank}(\mathbf{I} - \mathbf{P_X})$ and it follows that

$$\text{MSE} = \frac{\text{SSE}}{\text{n} - \text{k}}$$

is an unbiased estimator of σ^2.

Ans:

$\text{SSE} = \mathbf{Y^T}(\mathbf{I} - \mathbf{P_X})\mathbf{Y}$

$$\begin{aligned} E(SSE) &= \boldsymbol{\mu}^{\mathbf{T}}\mathbf{A}\boldsymbol{\mu} + tr(\mathbf{A}\boldsymbol{\Sigma}) \\ &= \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I} - P_{\mathbf{X}})\mathbf{X}\boldsymbol{\beta} tr[(\mathbf{I} - P_{\mathbf{X}})\sigma^2\mathbf{I} \\ &= \sigma^2 tr(\mathbf{I} - P_{\mathbf{X}}) \\ &= \sigma^2(n-k) \end{aligned}$$

$MSE = \frac{SSE}{n-k}$

$E[MSE] = \frac{\sigma^2(n-k)}{n-k} = \sigma^2$

$\therefore$ MSE is is unbiased estimator of σ^2.

(x) $\frac{1}{\sigma^2}SSE \sim \chi^2_{n-k}$

Ans:

$\frac{SSE}{\sigma^2} = \frac{\mathbf{Y}^T(\mathbf{I}-P_{\mathbf{X}})\mathbf{Y}}{\sigma^2}$

$A = \frac{\mathbf{I}-P_{\mathbf{X}}}{\sigma^2}$, $\Sigma = \sigma^2\mathbf{I}$, $A\Sigma = \mathbf{I} - P_{\mathbf{X}}$ is idempotent.

$d.f. = rank(\mathbf{I} - P_{\mathbf{X}}) = n - k$

$\lambda = \frac{1}{2}\boldsymbol{\beta}^T\mathbf{X}(\mathbf{I} - P_x)\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$

$\frac{SSE}{\sigma^2} \sim \chi^2_{n-k}$

(xi) SS_{H_0} and SSE are independently distributed.

Ans:

$\mathbf{Cb} = \mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}$, $\mathbf{e} = (\mathbf{I} - P_{\mathbf{X}})\mathbf{Y}$

$\begin{bmatrix} \mathbf{Cb} \\ \mathbf{e} \end{bmatrix} = \begin{bmatrix} \mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T \\ \mathbf{I} - P_{\mathbf{X}} \end{bmatrix}\mathbf{Y}$ has a multivariate normal distribution.

$Cov(\mathbf{Cb}, \mathbf{e})$
$= Cov[\mathbf{C}(\mathbf{X}T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}, (\mathbf{I} - P_{\mathbf{X}})\mathbf{Y}]$
$= \mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T V(\mathbf{Y})(\mathbf{I} - P_{\mathbf{X}})^T$
$= \sigma^2\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T(\mathbf{I} - P_{\mathbf{X}})$
$= \mathbf{0}$

$\therefore$ **Cb** and **e** are independent. Consequently SSH_0 is independent of SSE.

(xii) $F = \dfrac{\left(\frac{SSH_0}{m\sigma^2}\right)}{\left(\frac{SSE}{(n-k)\sigma^2}\right)} = \dfrac{\frac{SSH_0}{m}}{\frac{SSE}{n-k}} = \dfrac{(n-k)SSH_0}{mSSE}$

$\sim F_{m,n-k}(\lambda)$

with noncentrality parameter

$$\lambda = \frac{1}{\sigma^2}(\mathbf{C}\boldsymbol{\beta} - \mathbf{d})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T]^{-1}(\mathbf{C}\boldsymbol{\beta} - \mathbf{d})$$

$$\geq 0$$

and $\lambda = 0$ if and only if $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$ is true.

HT-Q01

Example 1.

Consider the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with

$$\mathbf{Y} = \begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \\ Y_{23} \\ Y_{24} \\ Y_{31} \\ Y_{32} \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{\beta} = \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$$

and $\boldsymbol{\epsilon} \sim N(0, \sigma^2 I)$.

(a) Determine which of the following hypotheses are testable.

i. $H_0 : \alpha_1 = \alpha_2$

Ans: $H_0 : \mathbf{c}^T\boldsymbol{\beta} = \begin{bmatrix} 0 & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = 0$

Let $\mathbf{a}^T = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, then $E(\mathbf{a}^T\mathbf{Y}) = \mathbf{c}^T\boldsymbol{\beta} \Rightarrow \mathbf{c}^T\boldsymbol{\beta}$ is estimable.

Rank$(\mathbf{c}^T) = 1 =$ number of row in $\mathbf{C}$, thus $H_0 : \alpha_1 = \alpha_2$ is testable.

ii. $H_0 : \alpha_1 - 2\alpha_2 + 3\alpha_3 = 0$

Ans:

$\mathbf{c}^T\boldsymbol{\beta} = \begin{bmatrix} 0 & 1 & -2 & 3 \end{bmatrix} \boldsymbol{\beta}$

Let $\mathbf{d}^T = \begin{bmatrix} 1 & -1 & -1 & -1 \end{bmatrix}$, then $\mathbf{Xd} = \mathbf{0}$ but $\mathbf{c}^T\mathbf{d} = -1 + 2 - 3 = -2 \neq 0$. Thus $\alpha_1 - 2\alpha_2 + 3\alpha_3$ is not estimable and hence $H_0 : \alpha_1 - 2\alpha_2 + 3\alpha_3 = 0$ is not testable.

iii. $H_0 : \alpha_3 = 0$

Ans:

$\mathbf{c}^T\boldsymbol{\beta} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \boldsymbol{\beta}$

Let $\mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$, then $\mathbf{Xd} = \mathbf{0}$ but $\mathbf{c}^T\mathbf{d} = -1 \neq 0$. Thus α_3 is not estimable and hence $H_0 : \alpha_3 = 0$ is not testable.

iv. $H_0 : \mu = 0$

Ans:

$\mathbf{c}^T\boldsymbol{\beta} = \begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix}\boldsymbol{\beta}$

Let $\mathbf{d} = \begin{bmatrix}1\\-1\\-1\\-1\end{bmatrix}$, then $\mathbf{Xd} = \mathbf{0}$ but $\mathbf{c}^T\mathbf{d} = 1 \neq 0$. Thus μ is not estimable and hence $H_0 : \mu = 0$ is not testable.

v. $\alpha_1 = \alpha_3$ and $\alpha_1 - 2\alpha_2 + \alpha_3 = 0$

Ans: Let $A = \begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0 & 0 & -1\\1 & 0 & -2 & 0 & 0 & 0 & 1 & 0\end{bmatrix}$, then

$E(A\mathbf{Y}) = \begin{bmatrix}\alpha_1 - \alpha_3\\\alpha_1 - 2\alpha_2 + \alpha_3\end{bmatrix}$

$\Rightarrow \mathbf{C}\boldsymbol{\beta}$ is estimable.

$Rank(\mathbf{C}) = 2 =$ rows of $\mathbf{C}$.

$\therefore H_0 : \mathbf{C}\boldsymbol{\beta} = 0$ is testable.

vi. $\alpha_1 = \alpha_3$ and $\alpha_2 = \alpha_3$ and $\alpha_1 + \alpha_2 - 2\alpha_3 = 0$

Ans: $\mathbf{C}\boldsymbol{\beta} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & -2 \end{bmatrix} \boldsymbol{\beta}$

$Rank(\mathbf{C}) = 2 \neq$ number of rows of $\mathbf{C}$. Thus $H_0 : \mathbf{C}\boldsymbol{\beta} = 0$ is not testable.

Example 2.

HT-Q48b

An experiment was conducted to study the effect of two diets (1 and 2) and two drugs (1 and 2) on blood pressure in rats. A total of 28 rats were randomly assigned to the 4 combinations of diet and drug, with 7 rats per combination. Let y_{ijk} be the decrease in blood pressure from the beginning of the study to the end of the study for diet i, drug j, and rat $k (i = 1, 2, j = 1, 2, k = 1, \ldots 7)$. Suppose

$$y_{ijk} = \mu_{ij} + \epsilon_{ijk}, (1)$$

where the μ_{ij} terms are unknown parameters and the ϵ_{ijk} terms are independent and identically distributed as $N(0, \sigma^2)$ for some unknown variance parameter$\sigma^2 > 0$. A researcher suspects the mean reduction in blood pressure will be the same for all combinations of diet and drug except for the combination of diet 1 with drug 2.This leads to consideration of the null hypothesis

$$H_0 : \mu_{11} = \mu_{21} = \mu_{22}.$$

Assuming model (1) holds,

(a) Write the hypothesis in the form $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$.

Ans:

$$H_0 : \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \mu_{11} \\ \mu_{12} \\ \mu_{21} \\ \mu_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

(b) Show that H_0 is testable.

Ans:
$E\begin{bmatrix} \bar{y}_{11} - \bar{y}_{21} \\ \bar{y}_{11} - \bar{y}_{22} \end{bmatrix} = \begin{bmatrix} \mu_{11} - \mu_{21} \\ \mu_{11} - \mu_{22} \end{bmatrix} = \mathbf{C}\boldsymbol{\beta} \Longrightarrow$
$\mathbf{C}\boldsymbol{\beta}$ is estimable.
$Rank(\mathbf{C}) = 2 =$ rows of $\mathbf{C}$.
$\therefore H_0 : \mathbf{C}\boldsymbol{\beta} = 0$ is testable.

(c) Determine the distribution of the F statistic you would use to test the null hypothesis and the degrees of freedom of the statistic;

Ans:
$F = \dfrac{SSH_0/df_{SSH_0}}{SSE/df_{SSE}} \sim F_{df_{SSH_0}, df_{SSE}}(\lambda)$
$df_{SSH_0} = Rank(\mathbf{C}) = 2$
$df_{SSE} = Rank(\mathbf{X}) - k = 28 - 4 = 24$

(d) Express the numerator of the F-statistic as a quadratic form.

Ans:

$SSH_0 = (\mathbf{Cb})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T]^{-1}(\mathbf{Cb})$

where

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} \hat{\mu}_{11} \\ \hat{\mu}_{12} \\ \hat{\mu}_{21} \\ \hat{\mu}_{22} \end{bmatrix} = \begin{bmatrix} \bar{y}_{11.} \\ \bar{y}_{12.} \\ \bar{y}_{21.} \\ \bar{y}_{22.} \end{bmatrix}; \mathbf{Cb} = \begin{bmatrix} \bar{y}_{11} - \bar{y}_{21} \\ \bar{y}_{11} - \bar{y}_{22} \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} \mathbf{1}_{7\times 1} & 0 & 0 & 0 \\ 0 & \mathbf{1}_{7\times 1} & 0 & 0 \\ 0 & 0 & \mathbf{1}_{7\times 1} & 0 \\ 0 & 0 & 0 & \mathbf{1}_{7\times 1} \end{bmatrix}; \mathbf{X^TX}^{-1} = \begin{bmatrix} \frac{1}{7} & 0 & 0 & 0 \\ 0 & \frac{1}{7} & 0 & 0 \\ 0 & 0 & \frac{1}{7} & 0 \\ 0 & 0 & 0 & \frac{1}{7} \end{bmatrix}$$

$$\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{C}^T = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{7} & 0 & 0 & 0 \\ 0 & \frac{1}{7} & 0 & 0 \\ 0 & 0 & \frac{1}{7} & 0 \\ 0 & 0 & 0 & \frac{1}{7} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} =$$

$$\frac{1}{7}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$[\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{C}^T]^{-1} = \frac{7}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\begin{aligned} SSH_0 &= \begin{bmatrix} \bar{y}_{11} - \bar{y}_{21} & \bar{y}_{11} - \bar{y}_{22} \end{bmatrix} \left(\frac{7}{3}\right) \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} \bar{y}_{11} - \bar{y}_{21} \\ \bar{y}_{11} - \bar{y}_{22} \end{bmatrix} \\ &= \frac{14}{3}\left[(\bar{y}_{11} - \bar{y}_{21})^2 + (\bar{y}_{11} - \bar{y}_{22})^2 - (\bar{y}_{11} - \bar{y}_{21})(\bar{y}_{11} - \bar{y}_{22})\right] \end{aligned}$$

(e) provide a fully simplified expression for the noncentrality parameter of the statistic in terms of model (1) parameters.

Ans:

Let

$$\boldsymbol{\beta} = \begin{bmatrix} \mu_{11} \\ \mu_{12} \\ \mu_{21} \\ \mu_{22} \end{bmatrix}$$

$\lambda = \frac{1}{\sigma^2}(\mathbf{C}\boldsymbol{\beta} - \mathbf{d})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T]^{-1}(\mathbf{C}\beta - \mathbf{d})$

$$\mathbf{C}\boldsymbol{\beta} - \mathbf{d} = \begin{bmatrix} \mu_{11} - \mu_{21} \\ \mu_{11} - \mu_{22} \end{bmatrix}$$

$$\begin{aligned} \lambda &= \frac{\begin{bmatrix} \mu_{11} - \mu_{21} & \mu_{11} - \mu_{22} \end{bmatrix} \left(\frac{7}{3}\right) \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} \mu_{11} - \mu_{21} \\ \mu_{11} - \mu_{22} \end{bmatrix}}{\sigma^2} \\ &= \frac{14}{3\sigma^2}\left[(\mu_{11} - \mu_{21})^2 + (\mu_{11} - \mu_{22})^2 - (\mu_{11} - \mu_{21})(\mu_{11} - \mu_{22})\right] \end{aligned}$$

Example 3.

HT-Q14b

An researcher recorded moisture content for three types of cheese made by two different methods. Two pieces of cheese were measure for each type and each method. The data are shown below.

Treatment	Moisture Content Measurements	
Type A made with Method 1	$y_{11} = 39.02$	$y_{12} = 38.79$
Type B made with Method 1	$y_{21} = 35.74$	$y_{22} = 35.41$
Type C made with Method 1	$y_{31} = 37.02$	$y_{32} = 36.0$
Type A made with Method 2	$y_{41} = 38.96$	$y_{42} = 39.01$
Type B made with Method 2	$y_{51} = 35.58$	$y_{52} = 35.52$
Type C made with Method 2	$y_{61} = 35.7$	$y_{62} = 36.0$

Consider the model $y_{ij} = \mu + \alpha_i + \epsilon_{ij}$, where $\epsilon_{ij} \sim NID(0, \sigma^2)$, $i = 1, 2, 3, 4, 5, 6$, and $j = 1, 2$. This model can be expressed in matrix form as $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$.

(a) Suppose the null hypothesis is:
"for each type of cheese, the average moisture content is not affected by the method for making cheese. (This hypothesis allows the average moisture content to vary across types of cheese)."
Express the hypothesis in the form $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{0}$.

Ans:
The hypothesis to be tested is

$$H_0 : \alpha_1 = \alpha_4; \alpha_2 = \alpha_5; \alpha_3 = \alpha_6$$

or

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \begin{bmatrix} 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 \end{bmatrix} \boldsymbol{\beta} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(b) Write down the $\mathbf{X}$ matrix in kronecker form.

Ans:
$\mathbf{X} = [\mathbf{1}_{12} | \mathbf{1}_2 \otimes \mathbf{I}_6]$

(c) Compute one of the solution to the normal equation, $\mathbf{b}$.

Ans:

$$\mathbf{X^T X} = \begin{bmatrix} 12 & 2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 \\ 2 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

Let $\mathbf{W} = 2\mathbf{I}_5$, then $\mathbf{W}^{-1} = \frac{1}{2}\mathbf{I}_5$, and $\mathbf{X^T X}^- = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$

$$\mathbf{X^Ty} = [\mathbf{1}_{12}|\mathbf{1}_2 \otimes \mathbf{I}_6]^T \begin{bmatrix} 39.02 \\ 38.79 \\ 35.74 \\ 35.41 \\ 37.02 \\ 36.0 \\ 38.96 \\ 39.01 \\ 35.58 \\ 35.52 \\ 35.7 \\ 36.0 \end{bmatrix} = \begin{bmatrix} 442.75 \\ 77.81 \\ 71.15 \\ 73.02 \\ 77.97 \\ 71.1 \\ 71.7 \end{bmatrix}$$

$$\mathbf{b} = (\mathbf{X^TX})^{-}\mathbf{X^Ty} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 442.75 \\ 77.81 \\ 71.15 \\ 73.02 \\ 77.97 \\ 71.1 \\ 71.7 \end{bmatrix} = \begin{bmatrix} 0 \\ 38.905 \\ 35.575 \\ 36.51 \\ 38.985 \\ 35.55 \\ 35.85 \end{bmatrix}$$

(d) Compute the SS_{H_0} corresponding to the null hypothesis in part (a), and state it's distribution when the null hypothesis is true.

Ans:

$$\mathbf{Cb} = \begin{bmatrix} 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 38.905 \\ 35.575 \\ 36.51 \\ 38.985 \\ 35.55 \\ 35.85 \end{bmatrix} = \begin{bmatrix} -0.08 \\ 0.025 \\ 0.66 \end{bmatrix}$$

;

$\mathbf{C}(\mathbf{X^TX})^{-}\mathbf{C}^T$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; (\mathbf{C}(\mathbf{X^TX})^{-}\mathbf{C}^T)^{-1} = \mathbf{I}_3$$

$$\begin{aligned} SSH_0 &= (\mathbf{Cb}-\mathbf{d})^{\mathbf{T}}(\mathbf{C}(\mathbf{X^TX})^{-}\mathbf{C^T})^{-1}(\mathbf{Cb}-\mathbf{d}) \\ &= \begin{bmatrix} -0.08 & 0.025 & 0.66 \end{bmatrix} \mathbf{I}_3 \begin{bmatrix} -0.08 \\ 0.025 \\ 0.66 \end{bmatrix} \\ &= 0.4426 \end{aligned}$$

Under H_0, $SSH_0 \sim \chi^2_3$

(e) Compute SSE.

Ans:

$$\hat{\mathbf{y}} = \begin{bmatrix} 38.905 \\ 38.905 \\ 35.575 \\ 35.575 \\ 36.51 \\ 36.51 \\ 38.985 \\ 38.985 \\ 35.55 \\ 35.55 \\ 35.85 \\ 35.85 \end{bmatrix}; \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 39.02 \\ 38.79 \\ 35.74 \\ 35.41 \\ 37.02 \\ 36.0 \\ 38.96 \\ 39.01 \\ 35.58 \\ 35.52 \\ 35.7 \\ 36.0 \end{bmatrix} - \begin{bmatrix} 38.905 \\ 38.905 \\ 35.575 \\ 35.575 \\ 36.51 \\ 36.51 \\ 38.985 \\ 38.985 \\ 35.55 \\ 35.55 \\ 35.85 \\ 35.85 \end{bmatrix} = \begin{bmatrix} 0.12 \\ -0.12 \\ 0.16 \\ -0.17 \\ 0.51 \\ -0.51 \\ -0.02 \\ 0.02 \\ 0.03 \\ -0.03 \\ -0.15 \\ 0.15 \end{bmatrix}$$

$$SSE = \mathbf{e^T e} = \begin{bmatrix} 0.12 \\ -0.12 \\ 0.16 \\ -0.17 \\ 0.51 \\ -0.51 \\ -0.02 \\ 0.02 \\ 0.03 \\ -0.03 \\ -0.15 \\ 0.15 \end{bmatrix}^T \begin{bmatrix} 0.12 \\ -0.12 \\ 0.16 \\ -0.17 \\ 0.51 \\ -0.51 \\ -0.02 \\ 0.02 \\ 0.03 \\ -0.03 \\ -0.15 \\ 0.15 \end{bmatrix} = 0.6511$$

(f) Compute the value of the corresponding $F-$statistic and report the degrees of freedom.

Ans:

$F = \frac{SSH_0/df_{H_0}}{SSE/df_{SSE}} = \frac{0.4426/3}{0.6511/6} = 1.3595$

$df = (3, 6)$

4.6 Elements of Hypothesis Test

We perform the test by rejecting

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$$

if

$$F > F_{(m,n-k),\alpha}$$

where α is a specified significance level (Type I error level) for the test.

$$\alpha = P\{rejectH_0 |\ H_0 \text{ is true}\}$$

4.6.1 Type I Error Level

$$\alpha = P\left\{F > F_{m,n-k,\alpha} \mid H_0 \text{ is true}\right\}$$

When H_0 is true,

$$F = \frac{MS_{H_0}}{MSE} \sim F_{m,n-k}$$

This is the probability of incorrectly rejecting a null hypothesis that is true.

4.6.2 Type II Error Level

$$\begin{aligned}\beta &= P\{\text{Type II error}\}\\ &= P\{\text{fail to reject } H_0 \mid H_0 \text{ is false}\}\\ &= P\{F < F_{m,n-k,\alpha} \mid H_0 \text{ is false}\}\end{aligned}$$

When H_0 is false,

$$F = \frac{MS_{H_0}}{MSE} \sim F_{(m,n-k)}(\lambda)$$

4.6.3 Power of a Test

$$\begin{aligned}power &= 1-\beta\\ &= P\{F > F_{m,n-k,\alpha} \mid H_0 \text{ is false}\}\end{aligned}$$

this determines the value of the noncentrality parameter.

For a fixed type I error level (significance level) α, the power of the test increases as the noncentrality parameter increases.

$$\lambda = \frac{1}{\sigma^2}(\mathbf{C}\boldsymbol{\beta}-\mathbf{d})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}^T]^{-1}(\mathbf{C}\boldsymbol{\beta}-\mathbf{d})$$

Example 4.

Effects of three diets on blood coagulation times in rats.

Diet factor: Diet 1, Diet 2, Diet 3

Response: blood coagulation time

Model for a completely randomized experiment with n_i rats assigned to the i-th diet.

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

where

$$\epsilon_{ij} \sim NID(0, \sigma^2)$$

for $i = 1, 2, 3$ and $j = 1, 2, ..., n_i$.

Here, $E(Y_{ij}) = \mu + \alpha_i$ is the mean coagulation time for rats fed the i-th diet.

Test the null hypothesis that the mean blood coagulation time is the same for all three diets.

Ans:

$H_0 : \mu + \alpha_1 = \mu + \alpha_2 = \mu + \alpha_3$

Equivalent ways to express the null hypothesis are:

$H_0 : \alpha_1 = \alpha_2 = \alpha_3$

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \begin{bmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Obtain the OLS estimator for $\mathbf{C}\boldsymbol{\beta}$,

$$\mathbf{Cb} = \mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}$$

and evaluate

$$\begin{aligned} SS_{H_0} &= (\mathbf{Cb} - \mathbf{0})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T]^{-1}(\mathbf{Cb} - \mathbf{0}) \\ &= \sum_{i=1}^{3} n_i(\bar{Y}_{i.} - \bar{Y}_{..})^2 \end{aligned}$$

$MS_{H_0} = \frac{SS_{H_0}}{2}$ on $3 - 1 = 2$ d.f.

$$SSE = \mathbf{Y}^T(I - P_{\mathbf{X}})\mathbf{Y} = \sum_{i=1}^{3}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_{i.})^2$$

$$MSE = \frac{SSE}{\sum_{i=1}^{3}(n_i - 1)} \quad \text{on } \sum_{\mathrm{i=1}}^{3}(\mathrm{n_i} - 1)\,\text{d.f.}$$

Reject H_0 in favor of H_1 if

$$F = \frac{\mathrm{MS_{H_0}}}{MSE} > F_{(2,\sum_{i=1}^{3}(n_i-1)),\alpha}$$

..

Example 5.

Suppose we are willing to specify:

(i) α = type I error level = .05

(ii) $n_1 = n_2 = n_3 = n$

(iii) power $\geq$ .90 to detect

(iv) a specific alternative

$$(\mu + \alpha_1) - (\mu + \alpha_3) = 0.5\sigma$$
$$(\mu + \alpha_2) - (\mu + \alpha_3) = \sigma$$

How many observations (in this case rats) are needed?

Ans: For this particular alternative

$$\mathbf{C}\boldsymbol{\beta} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} .5\sigma \\ \sigma \end{bmatrix}$$

and the power of the F-test is

$$\text{power} = \mathrm{P}\left\{\mathrm{F}_{(2,3\mathrm{n}-3)}(\lambda) > \mathrm{F}_{(2,3\mathrm{n}-3),.05}\right\}$$

where

$$\lambda = \frac{1}{\sigma^2}\left(\begin{bmatrix} .5\sigma \\ \sigma \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right)^T \left[\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T\right]^{-1}\left(\begin{bmatrix} .5\sigma \\ \sigma \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right)$$
$$= [.5\ 1]\left[\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T\right]^{-1}\begin{bmatrix} .5 \\ 1 \end{bmatrix}$$

$\mathbf{X}$ has n rows of (1 1 0 0)
n rows of (1 0 1 0)
n rows of (1 0 0 1)

In this case,

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 3n & n & n & n \\ n & n & 0 & 0 \\ n & 0 & n & 0 \\ n & 0 & 0 & n \end{bmatrix}$$

and a generalized inverse is

$$(\mathbf{X}^T\mathbf{X})^- = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & n^{-1} & 0 & 0 \\ 0 & 0 & n^{-1} & 0 \\ 0 & 0 & 0 & n^{-1} \end{bmatrix}$$

Then,

$$\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T = \frac{1}{n}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

and

$$\begin{aligned}\lambda &= [.5\ 1][\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T]^{-1}\begin{bmatrix} .5 \\ 1 \end{bmatrix} \\ &= \frac{n}{2}\end{aligned}$$

Choose n to achieve

$$\begin{aligned}.90 &= \text{power} \\ &= P\left\{F_{(2,3n-3)}\left(\frac{n}{2}\right) > F_{(2,3n-3).05}\right\}\end{aligned}$$

```
# This file is stored in powerR.txt

#==================================
```



```
n = 2:50
d2 =  n/2
power = 1 - pf(qf(.95,2,3*n-3),2,3*n-3,d2)

#  function here to open a graphics window

par(fin=c(7.5,8.5),cex=1.2,mex=1.5,lwd=4,
     mar=c(5.1,4.1,4.1,2.1))

plot((0,5), (0,1), type="n",
        xlab="Sample Size",ylab="Power",
main="Power versus Sample Size\n
        F-test for equal means")
lines(n, power, type="l",lty=1)



#=========================================
```

Example 6.

For the hypotheses testing

$$H_0 : (\mu + \alpha_1) = (\mu + \alpha_2) = \cdots = (\mu + \alpha_k)$$

against

$$H_1 : (\mu + \alpha_1) \neq (\mu + \alpha_j) \text{for some} i \neq j$$

Obtain the test statistic and the corresponding non-centrality parameter.

Ans:

use

$$F = \frac{MS_{H_0}}{MSE} \sim F_{(k-1, \sum_{i=1}^{k}(n_i - 1))}(\lambda)$$

where

$$\lambda = \frac{1}{\sigma^2} \sum_{i=1}^{k} n_i (\alpha_i - \bar{\alpha}_.)^2$$

with

$$\bar{\alpha}_. = \frac{\sum_{i=1}^{k} n_i \alpha_i}{\sum_{i=1}^{k} n_i}$$

To obtain the formula for the noncentrality parameter, write the null hypothesis as

$$H_0 : \mathbf{0} = \mathbf{C}\boldsymbol{\beta} = \left(\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & 0 & \ddots & \vdots & \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & \frac{n_1}{n_.} & \cdots & \frac{n_k}{n_.} \\ 0 & \vdots & & \\ \vdots & \vdots & & \\ 0 & \frac{n_1}{n_.} & \cdots & \frac{n_k}{n_.} \end{bmatrix} \right) \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_k \end{bmatrix}$$

Use

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 0 & & 0 \\ 1 & 1 & 0 & & 0 \\ 1 & 0 & 1 & & 0 \\ 1 & 0 & 1 & & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & 1 \\ \vdots & \vdots & \vdots & & 1 \\ 1 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} n_. & n_1 & n_2 & \cdots & n_k \\ n_1 & n_1 & & & \\ n_2 & & n_2 & & \\ \vdots & & & \ddots & \\ n_k & & & & n_k \end{bmatrix}$$

$$(\mathbf{X}^T\mathbf{X})^- = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & n_1^{-1} & 0 & \cdots & 0 \\ 0 & 0 & n_2^{-1} & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & \cdots & & n_k^{-1} \end{bmatrix}.$$

Then

$$\lambda = \frac{1}{\sigma^2}(\mathbf{C}\boldsymbol{\beta} - \mathbf{0})^T[\mathbf{C}(\mathbf{X}^T\mathbf{X})^-\mathbf{C}^T]^{-1}(\mathbf{C}\boldsymbol{\beta} - \mathbf{0})$$

$$= \frac{1}{\sigma^2}\sum_{i=1}^{k} n_i(\alpha_1 - \bar{\alpha}_.)^2$$

4.7 Confidence intervals for estimable functions of $\boldsymbol{\beta}$

Definition 2.

Suppose $Z \sim N(0,1)$ is distributed independently of $W \sim \chi^2_v$, and then the distribution of

$$t = \frac{Z}{\sqrt{\frac{W}{v}}}$$

is called the student t-distribution with v degrees of freedom. We will use the notation

$$t \sim t_v$$

For the normal-theory Gauss-Markov model

$$\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I),$$

the OLS estimator of an estimable function, $\mathbf{c}^T\boldsymbol{\beta}$,

$$\mathbf{c}^T\mathbf{b} = \mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T\mathbf{Y}$$

follows a normal distribution, i.e.,

$$\mathbf{c}^T\mathbf{b} \sim N(\mathbf{c}^T\boldsymbol{\beta}, \sigma^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{c}).$$

It follows that

$$Z = \frac{\mathbf{c}^T\mathbf{b} - \mathbf{c}^T\boldsymbol{\beta}}{\sqrt{\sigma^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{c}}} \sim N(0,1)$$

From Result 1.(ix), we have

$$\frac{1}{\sigma^2}SSE = \frac{1}{\sigma^2}\mathbf{Y}^T(I - P_{\mathbf{X}})\mathbf{Y} \sim \chi^2_{(n-k)}$$

where $k = \text{rank}(\mathbf{X})$.

Using the same argument that we used to derive Result 1.(x), we can show that $c^T\mathbf{b}$ is distributed independently of $\frac{1}{\sigma^2}$ SSE.

First note that

$$\begin{bmatrix} \mathbf{c}^T\mathbf{b} \\ (I - P_{\mathbf{X}})\mathbf{Y} \end{bmatrix} \begin{bmatrix} \mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T \\ (I - P_{\mathbf{X}}) \end{bmatrix} \mathbf{Y}$$

has a joint normal distribution under the normal-theory Gauss-Markov model.

Note that

$$\begin{aligned} &Cov(\mathbf{c}^T\mathbf{b}, (I - P_{\mathbf{X}})\mathbf{Y}) \\ &\quad = (\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T)(V(\mathbf{Y}))(I - P_{\mathbf{X}})^T \\ &\quad = (\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{X}^T)(\sigma^2)(I - P_{\mathbf{X}}) \\ &\quad = \sigma^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\underbrace{\mathbf{X}^{\mathbf{T}}(\mathbf{I} - \mathbf{P}_{\mathbf{X}})}_{\uparrow} \\ &\quad = 0 \end{aligned}$$

this is a matrix of zeros

Consequently,
$\mathbf{c}^T\mathbf{b}$ is distributed independently of
$\mathbf{e} = (I - P_{\mathbf{X}})\mathbf{Y}$

which implies that

$\mathbf{c}^T\mathbf{b}$ is distributed independently of SSE $= \mathbf{e}^T\mathbf{e}$.

Then,

$$t = \frac{Z}{\sqrt{\frac{\text{SSE}}{\sigma^2(n-k)}}}$$

$$= \frac{\frac{\mathbf{c}^T\mathbf{b}-\mathbf{c}^T\boldsymbol{\beta}}{\sqrt{\sigma^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{c}}}}{\sqrt{\frac{\text{SSE}}{\sigma^2(n-k)}}}$$

$$= \frac{\mathbf{c}^T\mathbf{b}-\mathbf{c}^T\boldsymbol{\beta}}{\sqrt{\frac{\text{SSE}}{(n-k)}\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^-\mathbf{c}}} \sim t_{(n-k)}$$

$\frac{\text{SSE}}{n-k}$ is the MSE

It follows that

$$\begin{aligned} 1-\alpha &= P\left\{-t_{(n-k),\alpha/2} \leq \frac{\mathbf{c}^T\mathbf{b}-\mathbf{c}^T\boldsymbol{\beta}}{\sqrt{\text{MSE}\,\mathbf{c}^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^-\mathbf{c}}} \leq t_{(n-k),\alpha/2}\right\} \\ &= P\left\{\mathbf{c}^T\mathbf{b} - t_{(n-k),\alpha/2}\sqrt{\text{MSE}\,\mathbf{c}^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^-\mathbf{c}} \leq \mathbf{c}^T\boldsymbol{\beta}\right. \\ &\qquad \left. \leq \mathbf{c}^T\mathbf{b} + t_{(n-k),\alpha/2}\sqrt{\text{MSE}\,\mathbf{c}^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^-\mathbf{c}}\right\} \end{aligned}$$

and a $(1-\alpha)\times 100\%$ confidence interval for $\mathbf{c}^T\boldsymbol{\beta}$ is

$$\Big(\mathbf{c}^T\mathbf{b} - t_{(n-k),\alpha/2}\sqrt{\text{MSE}\,\mathbf{c}^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^-\mathbf{c}}\,,$$
$$\mathbf{c}^T\mathbf{b} + t_{(n-k),\alpha/2}\sqrt{\text{MSE}\,\mathbf{c}^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^-\mathbf{c}}\Big)$$

For brevity we will also write

$$\mathbf{c}^T\mathbf{b} \pm t_{(n-k),\alpha/2}\ S_{\mathbf{c}^T\mathbf{b}}$$

where

$$S_{\mathbf{c}^T\mathbf{b}} = \sqrt{\text{MSE}\,\mathbf{c}^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^-\mathbf{c}}\,.$$

For the normal-theory Gauss-Markov model with $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I)$, the interval

$$\mathbf{c}^T\mathbf{b} \pm t_{(n-k),\alpha/2}S_{\mathbf{c}^T\mathbf{b}}$$

is the **shortest random interval** with probability $(1-\alpha)$ of containing $\mathbf{c}^T\boldsymbol{\beta}$.

4.8 Confidence interval for σ^2:

For the normal-theory Gauss-Markov model with $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I)$ we have shown that

$$\frac{\text{SSE}}{\sigma^2} = \frac{\mathbf{Y}^T(I - P_{\mathbf{X}})\mathbf{Y}}{\sigma^2} \sim \chi^2_{(n-k)}$$

Then,

$$\begin{aligned} 1-\alpha &= P\left\{\chi^2_{(n-k),1-\alpha/2} \le \frac{\text{SSE}}{\sigma^2} \le \chi_{(n-k),\alpha/2}\right\} \\ &= P\left\{\frac{\text{SSE}}{\chi^2_{(n-k),\alpha/2}} \le \sigma^2 \le \frac{\text{SSE}}{\chi_{(n-k),1-\alpha/2}}\right\} \end{aligned}$$

The resulting $(1-\alpha)\times 100\%$ confidence interval for σ^2 is

$$\left(\frac{\text{SSE}}{\chi^2_{(n-k),\alpha/2}}, \frac{\text{SSE}}{\chi^2_{(n-k),1-\alpha/2}}\right)$$

Example 7.

For the simple regression model

$$Y_i = \beta_0 + \beta_1 \mathbf{X}_{i1} + \epsilon_i,$$

where for $\mathbf{e} = (\epsilon_1, \ldots, \epsilon_i)^T$, $E(\mathbf{e}) = \mathbf{0}$. You are given

x	2	3	4	5
y	4	7	6	8

Suppose that $V(\mathbf{e}) = \sigma^2 I$.

(a) Construct the 95% confidence interval for β_1.

Ans:

$$\mathbf{X} = \begin{bmatrix}1 & 2\\1 & 3\\1 & 4\\1 & 5\end{bmatrix}; \mathbf{X}^T\mathbf{X} = \begin{bmatrix}4 & 14\\14 & 54\end{bmatrix};$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{20}\begin{bmatrix}54 & -14\\-14 & 4\end{bmatrix} = \begin{bmatrix}2.7 & -0.7\\-0.7 & 0.2\end{bmatrix}$$

$$\mathbf{X}^T\mathbf{Y} = \begin{bmatrix}1 & 1 & 1 & 1\\2 & 3 & 4 & 5\end{bmatrix}\begin{bmatrix}4\\7\\6\\8\end{bmatrix} = \begin{bmatrix}25\\93\end{bmatrix}$$

$\mathbf{b} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} = \begin{bmatrix} 2.7 & -0.7 \\ -0.7 & 0.2 \end{bmatrix} \begin{bmatrix} 25 \\ 93 \end{bmatrix} = \begin{bmatrix} 2.4 \\ 1.1 \end{bmatrix}$

$\hat{\mathbf{Y}} = \mathbf{Xb} = \begin{bmatrix} 4.6 \\ 5.7 \\ 6.8 \\ 7.9 \end{bmatrix}$

$\hat{\mathbf{e}} = \mathbf{Y} - \hat{\mathbf{Y}} = \begin{bmatrix} 4 \\ 7 \\ 6 \\ 8 \end{bmatrix} - \begin{bmatrix} 4.6 \\ 5.7 \\ 6.8 \\ 7.9 \end{bmatrix} = \begin{bmatrix} -0.6 \\ 1.3 \\ -0.8 \\ 0.1 \end{bmatrix}$

$SSE = \hat{\mathbf{e}}^{\mathbf{T}}\hat{\mathbf{e}} = 2.7$

$MSE = \frac{2.7}{4-2} = 1.35$

$\mathbf{C}^T(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 2.7 & -0.7 \\ -0.7 & 0.2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0.2$

$t_2(.025) = qt(.975, 2) = 4.3$

95% confidence interval for β_1 is

$1.1 \pm t_2(.025)\sqrt{MSE\mathbf{C}^T(\mathbf{X}^T\mathbf{X})^{-}\mathbf{C}}$

$= 1.1 \pm 4.3\sqrt{1.35(0.2)}$

$= (-1.1343, 3.3343)$

(b) Give 95% two-sided confidence interval for σ^2 in the normal version model.

Ans:

$\chi^2_2(.025) = qchisq(.975, 2) = 7.38$

$\chi^2_2(.925) = qchisq(.025, 2) = 0.05$

95% two-sided confidence interval for σ^2 is

$\left(\frac{2.7}{7.38}, \frac{2.7}{.05}\right)$

$= (0.366, 54)$